

Internship Report

Score Matching techniques on Goodness-of-Fit tests

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Goodness-of-Fit Testing High-dimensional data and images KSD requires the score function

$$s_q(x) = \nabla_x \log q(x)$$

Learn the score using Score Matching

Objectives

- ▶ Evaluate KSD-U on toy Gaussian distributions.
- ▶ Combine Score Matching with KSD for MNIST.
- ▶ Generate samples using Diffusion Posterior Sampling.
- ▶ Apply KSD to the generated samples.

Goodness-of-Fit problem:

$$H_0 : p = q \quad \text{vs.} \quad H_1 : p \neq q$$

Given samples $x_i \sim p(x)$ and the reference score

$$s_q(x) = \nabla_x \log q(x),$$

KSD measures the discrepancy between the unknown distribution p and the reference distribution q .

KSD-U statistic:

$$\hat{S}_u(p, q) = \frac{1}{n(n-1)} \sum_{i \neq j} u_q(x_i, x_j)$$

where $u_q(x_i, x_j)$ is the kernelized Stein kernel.

where

$$u_q(x, x') = s_q(x)^\top k(x, x') s_q(x') + s_q(x)^\top \nabla_{x'} k(x, x') \\ + \nabla_x k(x, x')^\top s_q(x') + \text{tr}(\nabla_{x, x'} k(x, x')) .$$

RBF kernel:

$$k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2h^2}\right), \quad \boxed{h = \sqrt{\frac{D}{2}}}$$

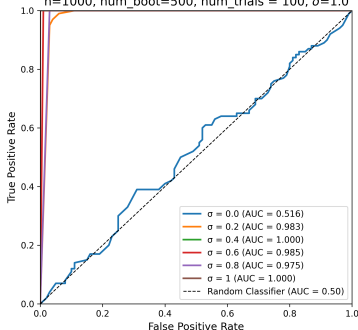
Hypothesis test: Bootstrap the KSD-U statistic and reject H_0 when the observed statistic is sufficiently large.

Experimental setup

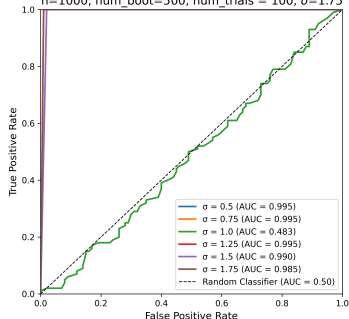
$$H_0 : X \sim \mathcal{N}(0, \sigma^2) \quad \text{vs.} \quad H_1 : X \sim \mathcal{N}(\delta, \sigma^2)$$

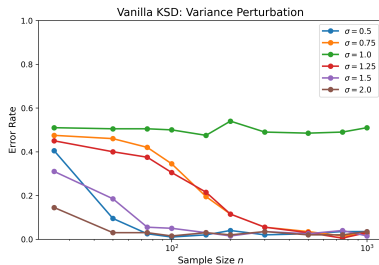
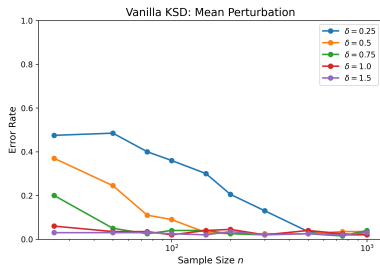
- ▶ Vary the mean δ or variance σ of the alternative.
- ▶ Evaluate the KSD-U bootstrap test using ROC, AUC and error rate.

Vanilla KSD ROC Comparison Null vs Alt with different means
n=1000, num_boot=500, num_trials = 100, $\delta=1.0$



Vanilla KSD ROC Comparison Null vs Alt with different variances
n=1000, num_boot=500, num_trials = 100, $\sigma=1.75$





Key observations

- ▶ High AUC for sufficiently large distributional perturbations.
- ▶ Error rate decreases with increasing sample size.
- ▶ Performance is close to random when the null and alternative coincide.

Why learn the score? Score is difficult to compute analytically for complex image distributions.

Score Matching

$$x_0 \xrightarrow{\text{add noise}} x_t = x_0 + \sigma_t z \xrightarrow{\text{score network}} s_\theta(x_t, t)$$

where

$$z \sim \mathcal{N}(0, I).$$

The conditional score is available analytically:

$$\nabla_{x_t} \log q_t(x_t | x_0) = -\frac{z}{\sigma_t}$$

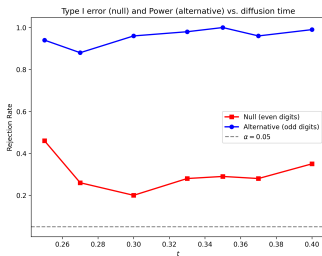
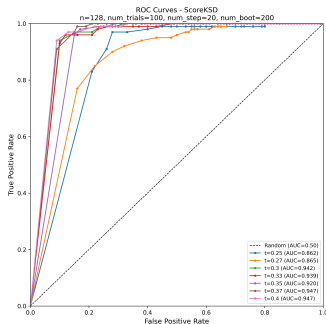
The network is trained using the score-matching objective:

$$\theta^* = \arg \min_{\theta} \mathbb{E} \left[\sigma_t \left\| s_\theta(x_t, t) + \frac{z}{\sigma_t} \right\|_2^2 \right]$$

Experiment:

H_0 : Even MNIST vs. H_1 : Odd MNIST

- ▶ Score network trained on MNIST even digits.
- ▶ $\sigma = 25$, with diffusion time t tuned experimentally.
- ▶ 128 samples, 200 bootstrap samples, 100 trials.



Best AUC ≈ 0.947

Key issue: limited statistical power

Inverse problem

$$y = A(x_0) + \epsilon$$

Noisy observation y $\longrightarrow$ Posterior samples x_0

DPS with the learned score

1. Start from noisy sample x_t .
2. Estimate the clean image using Tweedie's formula:

$$\hat{x}_0 = x_t + \sigma_t^2 s_\theta(x_t, t).$$

3. Measure consistency with the observation:

$$L_{\text{meas}} = \|y - A(\hat{x}_0)\|_2^2.$$

4. Correct the reverse diffusion step:

$$x_{t-dt} = x'_{t-dt} - \zeta_t \nabla_{x_t} L_{\text{meas}}.$$

Reverse SDE + measurement correction

Reconstruction settings

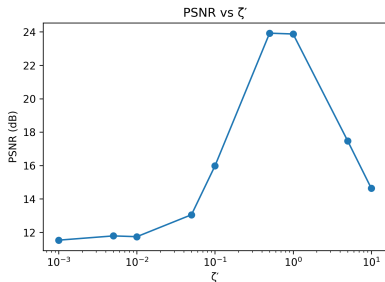
► Gaussian noise

► 20% Inpainting

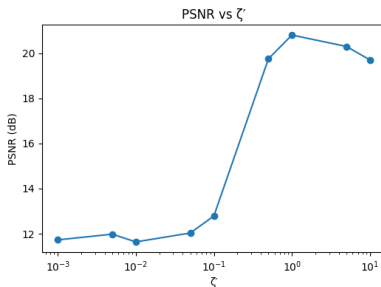
► Guidance parameter:

$$\zeta' \in \{0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1, 5, 10\}$$

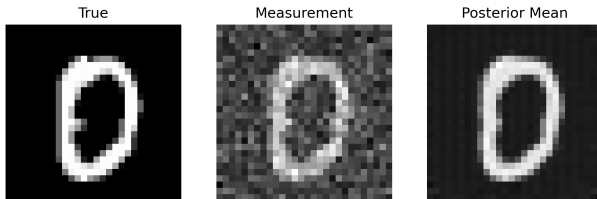
Gaussian



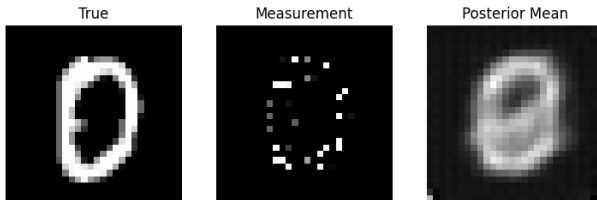
Inpainting



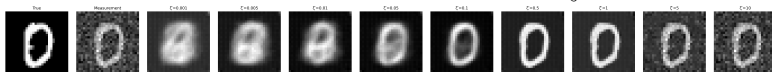
Gaussian Noise



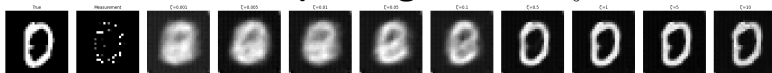
20% Inpainting



Gaussian Noise Different ζ'

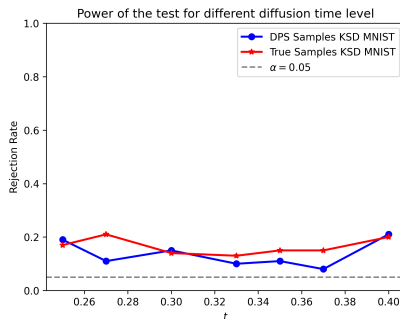


20% Inpainting Different ζ'



Goal: Test whether the generated DPS samples retain the statistical properties required by the KSD framework.

KSD Power Test



- ▶ Compare KSD power for true samples and DPS samples.
- ▶ The overall power is similar for both cases.
- ▶ Statistical power is still below the desired level.

What we achieved

1. **KSD-U** Effective goodness-of-fit testing on Gaussian toy examples.
2. **Learned Score + KSD** Extended KSD to high-dimensional MNIST data using Score Matching.
3. **DPS** Generated posterior samples for Gaussian noise and inpainting.
4. **KSD + DPS** KSD power on DPS samples is comparable to that on true samples.

Future Work

- ▶ Learned kernels for improved statistical power.
- ▶ Perturbed kernels / Pooled KSD for multimodal settings.
- ▶ Comparison with MMD and other GoF methods.

Thank You

Questions?